



PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 2

Introduction to Programming

SESSION / ASSIGNMENT – 12

TASKS :

1. Declare a structure named Playlist to store details of a song: title (string), artist (string), and duration in seconds (integer). Initialize one Playlist variable with your favorite song's details and print each field.



```
#include <stdio.h>

struct Playlist
{
    char title[50];
    char artist[50];
    int duration;
};

int main()
{
    struct Playlist song = {"Believer", "Imagine Dragons", 204};

    printf("Title: %s\n", song.title);
    printf("Artist: %s\n", song.artist);
    printf("Duration: %d seconds\n", song.duration);

    return 0;
}
```

OUTPUT:

```
Title: Believer
Artist: Imagine Dragons
Duration: 204 seconds
```

2. Create a structure called `FoodItem` to store Zomato-style menu data: `itemName` (string), `price` (float), and `rating` (float). Initialize an array of 3 `FoodItem` variables with real menu items and display their details using a loop.



```
#include <stdio.h>

struct FoodItem
{
    char itemName[30];
    float price;
    float rating;
};

int main()
{
    struct FoodItem menu[3] =
    {
        {"Veg Burger", 120.0, 4.5},
        {"Pizza", 299.0, 4.7},
        {"Cold Coffee", 99.0, 4.3}
    };

    int i;

    printf("Zomato Menu:\n\n");

    for(i = 0; i < 3; i++)
    {
        printf("Item: %s\n", menu[i].itemName);
        printf("Price: %.2f\n", menu[i].price);
        printf("Rating: %.1f\n\n", menu[i].rating);
    }

    return 0;
}
```

OUTPUT:

```
Zomato Menu:

Item: Veg Burger
Price: 120.00
Rating: 4.5

Item: Pizza
Price: 299.00
Rating: 4.7

Item: Cold Coffee
Price: 99.00
Rating: 4.3
```

3. Define a nested structure called MovieShow for a BookMyShow-style app: Movie (string), Screen (integer), and a nested structure Time with hours and minutes (integers). Create and initialize a MovieShow variable for any movie and print its details in the format 'Movie: X, Screen: Y, Time: HH:MM'.



```
#include <stdio.h>

struct Time
{
    int hours;
    int minutes;
};

struct MovieShow
{
    char movie[50];
    int screen;
    struct Time showTime;
};

int main()
{
    struct MovieShow show = {"Avengers: Endgame", 3, {7, 30}};

    printf("Movie: %s\n", show.movie);
    printf("Screen: %d\n", show.screen);
    printf("Time: %02d:%02d\n", show.showTime.hours, show.showTime.minutes);

    return 0;
}
```

OUTPUT:

```
Movie: Avengers: Endgame
Screen: 3
Time: 07:30
```

4. Build a structure called InstaProfile with fields: username (string), followers (integer), and a nested structure Bio with fields: description (string) and age (integer). Initialize an InstaProfile variable with your own details and display all fields.



```
#include <stdio.h>

struct Bio
{
    char description[100];
    int age;
};

struct InstaProfile
{
    char username[30];
    int followers;
    struct Bio bio;
};

int main()
{
    struct InstaProfile profile =
    {
        "haaris_ajmeri",
        850,
        {"B.Tech IT Student | Pyhton Full Stack Learner", 23}
    };

    printf("Username: %s\n", profile.username);
    printf("Followers: %d\n", profile.followers);
    printf("Bio: %s\n", profile.bio.description);
    printf("Age: %d\n", profile.bio.age);

    return 0;
}
```

OUTPUT:

```
Username: haaris_ajmeri
Followers: 850
Bio: B.Tech IT Student | Pyhton Full Stack Learner
Age: 23
```